Assignment # 1

DBMS

Total Marks: 50
Submission Date: 18 March 2026 10:00 am

Name__________________________ Roll No. _______________________ Class: ___________

Instructions:

- **Group Formation**
  - Each group must consist of 5 members only.
  - Students must submit the names and roll numbers of all group members on the title page
  - Work should be equally divided among all group members.
- Assignment **should be hand written.** (Neat and Clean A4 size paper with title page)
- You should create your own solution according to your logic. Any similarity with other student and internet will be consider as cheating and your assignment will be marked as ZERO.

- **Submission Guidelines**
  - Submit one assignment per group on portal.
  - Attach this page as front page of your assignment solution.
  - The assignment must be submitted before the deadline.
  - Late submissions may receive grade deductions.

For Each of the following Case Studies,

i. Identifying Entities
ii. Find relevant Attributes identify the attribute types.
iii. Identify the relationships between entities and determine the cardinality and participation of the relationships.
iv. Reduce the ERD discovered in step (iii) to Relational Schema. Show all relevant ERD changes during the process.

Case Study 1

A new International company want to open online grocery store in Pakistan. They decided to first start with 3 big cities of Pakistan (Karachi, Lahore and Islamabad) and start their services in these cities. Their business modal is same as “my cart” and working structure is also same. You are required to design the database. For this purpose, you need define the Entities and their required attributes. The attributes that work as keys and each entity cardinality.

Draw the ER-Diagram and convert in to Relational Model. You should also perform all DDL queries to convert your logical schema into physical schema and also suggest a suitable name for the domain.

Case Study 2

A store wants to keep track of all its musical albums.

1. Musical album have a title, price , a unique identifier as albumId
2. There are artists who create albums. We save artists name and CNIC for our record.
3. One artist can create multiple albums. But one album can only belong to a single artist.

4. One album can be of a single genre.
5. Genre has genereID , name, description.
6. Multiple albums can have a same genre.

Case Study 3

Draw an entity relationship diagram for the given senario. Mention keys, cardinality ratios and participation with proper notation. State assumption if any.

- . Consider a MOVIEdatabase in which data is recorded about the movie industry.The data requirements are summarized as follows:
- Each movie is identified by title and year of release. Each movie has a length in minutes. Each has a production company, and each is classified under one or more genres (such as horror, action, drama, and so forth). Each movie has one or more directors and one or more actors appear in it.Each movie also has a plot outline.Finally,each movie has zero or more quotable quotes, each of which is spoken by a particular actor appearing in the movie.
- Actors are identified by name and date of birth and appear in one or more movies.Each actor has a role in the movie.
- Directors are also identified by name and date of birth and direct one or more movies.It is possible for a director to act in a movie (including one that he or she may also direct).
- Production companies are identified by name and each has an address.A production company produces one or more movies.

Case Study 4

- The company is organized into departments. Each department has a unique name, a unique number, and a particular employee who manages the department. We keep track of the start date when that employee began managing the department. A department may have several locations.
- A department controls a number of projects, each of which has a unique name, a unique number, and a single location.
- We store each employee's name, Social Security number,2 address, salary, gender , and birth date. An employee is assigned to one department, but may work on several projects, which are not necessarily controlled by the same department. We keep track of the current number of hours per week that an employee works on each project. We also keep track of the direct supervisor of each employee (who is another employee).
- We want to keep track of the dependents of each employee for insurance purposes. We keep each dependent's first name, gender, birth date, and relationship to the employee.

Case Study 5

Consider a library management system. Library system keep tracks of books. For a book it is necessary that its ID and title should be stored in the database. A publisher publishes many books so their names should be kept in the records so that we can keep track of books and their publishers. Books are written by authors. Author related information is of no interest to library so only its name is saved in the database; nothing else. For publishers there must be address, phone number and their names saved in the database. The library has certain branches countrywide. These branches are differentiated using branch ids. Definitely each branch will have their own address and name. A person comes to library borrows a book against the card number that has been issued to him. All the personal details of the person will be saved against that card number. He can borrow it from any branch of the library by showing his card over there. Also the library system needs to maintain for a person that when he issued a certain book and when the due date is to return the book back to library. In addition to this a library branch can have multiple copies of the same book. It has to be maintained as well so that one can know how many copies we are left with in the library branch.

Case Study 6

Consider designing an ER for a university system. There are students' departments courses involved in this system. Student has student id, name address degree dob and age stored against them. In addition to this we want to store address as street address city country. A student does his majors in some department. Department has their code name office number. Department offers multiple courses. Course has name, description course number and credit hours. A course can be taught in multiple sections. Each section has section number and semester year and an instructor associated with it. Students study a course in a section and get some grades. Their grades and marks and gpa need to be stored in the database too.